

Multimodal Deep Learning

Answer Sheet 2

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Exercise 1: Numerical Features: Normalization vs. Standardization

A one-dimensional sensor produces readings

$$\mathbf{x} = [2.7, 3.1, 9.4, 6.5, 4.2, 3.8].$$

- (c) When would you prefer Min–Max scaling over z-scaling, and vice-versa?

Answer: Min-Max scaling is preferred when the true scale of information are sensitive to the quality of the output. While **z-scaling** is preferred when the input is approximately Gaussian, and the distribution of features can enhance the quality of the output

- (d) **Binning / Discretization:** Why convert a continuous feature (e.g. “age”) into bins (“young”, “middle-aged”, “senior”)? Give one benefit and one drawback.

Answer: One good benefit is discretization can help group a continuous feature into a simpler format which also help reduce the memory consumption, while one drawback is its lack of feature awareness and ruin system quality, particularly when binning algorithm is too naive or trying to do without sufficient data understanding.

Exercise 2: Categorical Features

You have a categorical attribute `Color` with values

$$[\text{Red}, \text{Blue}, \text{Green}, \text{Blue}].$$

- (a) Show the One-Hot encoded matrix.

Answer:

$$\mathbf{X} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

- (b) Show the Label-Encoding (integers assigned alphabetically from 0).

Answer:

$$\text{Blue} \mapsto 0, \text{Green} \mapsto 1, \text{Red} \mapsto 2$$

- (c) What can go wrong if you feed the labels from (b) directly into a linear or distance-based model?

Answer: The model will not be able to achieve its best performance because the label-encoding method does not manifest the exact relationship between each color. However, One-Hot encoding is a better option because it presents the vectors of color in vector space avoiding assumption of order or magnitude.

Exercise 3: Text Features

Byte-pair encoding (BPE) corpus

["low", "lower", "lowest", "newest"]

Perform 4 BPE merges. Show the merge pairs and the updated vocabulary after each step.

Answer: Please see the answer provided by the lecturer.

Exercise 4: Johnson–Lindenstrauss Lemma Bounds

For simplicity in this exercise, we use the bound $C = 8$.

- (a) $N = 10^6$, $\epsilon = 0.1$: compute the minimum k .

$$k \geq \frac{C \log N}{\epsilon^2} = \frac{8 \log 10^6}{0.1^2} \approx 11052.$$

Minimum $k = 4800$.

- (b) With $k = 500$ and $N = 10^6$, what is the best guaranteed distortion ϵ ?

$$\epsilon \geq \sqrt{\frac{C \log N}{k}} = \sqrt{\frac{8 \log 10^6}{500}} \approx 0.47.$$

The best guaranteed distortion $\epsilon = 0.47 = 47\%$.

Exercise 5: Sparse vs. Dense Distance Functions

- (a) **Sparse binary vectors:**

Answer: Euclidean distance cannot be utilized effectively with sparse data because the 0 and 1 values do not carry meaningful information for calculating distance, unlike in dense data, where features tend to have more varied and informative values.

- (b) **Unit-normalized dense vectors:**

Answer: From the provided answer, ranking values of $s \mapsto d(u, v) = \sqrt{2 - 2s}$ by both of them are equivalent after $L2$ -norm. This can refer that for certain tasks which require the order of values, we can effectively approximate the sequence of data using Euclidean distance – lower computational cost (e.g., Approximate Nearest Neighbor (ANN) Search)

- (c) **Dense count vectors of different length:**

Answer: Cosine similarity is invariant to the length of the document, unlike Euclidean distance.

Exercise 6: PCA via SVD

Given the mean-centred matrix

$$\mathbf{X} = \begin{bmatrix} -1 & 2 & 0 \\ -1 & -3 & 1 \\ 2 & 1 & -1 \end{bmatrix}$$

perform the singular-value decomposition $\mathbf{X} = U\Sigma V^T$. (Feel free to use a linear-algebra library such as ‘numpy.linalg.svd’.)

- (a) Give
- U
- ,
- Σ
- , and,
- V^T

Answer:

$$U = \begin{bmatrix} -0.3518 & 0.7368 & 0.5774 \\ 0.8140 & -0.0637 & 0.5774 \\ -0.4622 & -0.6731 & 0.5774 \end{bmatrix}, \Sigma = \begin{bmatrix} 4.07035 & 0 & 0 \\ 0 & 2.3307 & 0 \\ 0 & 0 & 0 \end{bmatrix}, V = \begin{bmatrix} -0.3406 & -0.8664 & 0.3651 \\ -0.8864 & 0.4254 & 0.1826 \\ 0.3135 & 0.2615 & 0.9129 \end{bmatrix}$$

- (b) Using the first principal component, form the rank-1 reconstruction
- $\mathbf{X}_1$
- , compute the reconstruction error
- $\|\mathbf{X} - \mathbf{X}_1\|_F$
- using the Frobenius norm.

Answer:

$$\begin{aligned} \mathbf{X}_1 &= \sigma_1 u_1 v_1^T \\ &= 4.07035 \cdot \begin{bmatrix} -0.3518 \\ 0.8140 \\ -0.4622 \end{bmatrix} \cdot [-0.3406 \quad -0.8864 \quad 0.3135] \\ &= \begin{bmatrix} 0.4876 & -1.2706 & 0.4497 \\ -1.1291 & -2.9383 & 1.041 \\ 0.6407 & 1.6670 & -0.5905 \end{bmatrix} \end{aligned}$$

$$\|\mathbf{X} - \mathbf{X}_1\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |x_{ij}|^2}$$

- (c) For a general data set, how can we choose
- k
- so that the reconstruction retains at least
- α
- fraction of the total variance?

Answer:

$$\frac{\sum_{i=1}^k \sigma_i^2}{\sum_{i=1}^r \sigma_i^2} \geq \alpha$$

where σ_i are the singular values of the data matrix $\mathbf{X}$, and r is the rank of $\mathbf{X}$. This ensures that the rank- k approximation captures at least $\alpha \times 100\%$ of the total variance in the data.

- (d) Relate principal components to
- intrinsic dimension*
- .

Answer: The principal components of a dataset, obtained via PCA, represent directions of maximal variance. The intrinsic dimension of the data corresponds to the smallest number of principal components required to capture most of the variance – typically 90-99%. This means the intrinsic dimension can be estimated by counting the number of significant singular values (or principal components), reflecting the true degrees of freedom in the data despite a possibly higher ambient dimension.